Name:_____ Class:_____ Student number:_____

Read only after completing B. Keep A available; check these answers without overwriting your first responses.

B1: labels, variables and fair comparisons

Labels: A uncovered; B tip shaded; C shaded below the tip.

Independent variable	Light-blocking treatment
Dependent variable	The coleoptile（胚芽鞘）’s bending response
Controlled variables	Light direction and intensity; Coleoptile（胚芽鞘） type, starting height and growth stage; Time exposed to light; Temperature and water supply

Valid assumptions: similar initial growth condition; shading does not restrict elongation（延長） or bending; shading creates no clear temperature difference. The same species does not remove the need to control initial size.

A is uncovered and provides a baseline without shading in the same environment. A–B changes tip shading; A–C changes shading below the tip. Keep conditions such as initial size and temperature the same to avoid other influences.

B2 / B3 / B4: check observations

Setup	Treatment	Direction
A	No light blocked	Bends left
B	Light blocked at the tip	No clear bending
C	Light blocked below the tip	Bends left

Setup	Treatment	Elongation（延長）
A	No light blocked	Yes
D	Tip removed	No

Setup	Treatment	Angle	Direction
E	Blank agar（瓊脂） • Centre	0°（Teaching model）	No clear bending
F	Treated agar（瓊脂） • Centre	0°（Teaching model）	No clear bending
G	Treated agar（瓊脂） • Left	35°（Teaching model; ±2°）	Bends right
H	Treated agar（瓊脂） • Right	35°（Teaching model; ±2°）	Bends left

Method check: E/F are 0°. G/H accept 33–37°, with equal angle magnitudes but opposite directions. Observations and readings are model data, not auxin（生長素） concentrations.

B2 / B3 / B4: check analysis

3. Which part may sense light from one side? Tip

1. Compare A and B: What changes when light is blocked at the tip? No clear bending towards the light.

2. Compare A and C: Does it still bend when the lower part receives no direct light? It can still bend towards light.

1. What does comparing elongation（延長） in A and D support? The presence of the tip also affects the coleoptile（胚芽鞘）’s growth / elongation（延長）.

1. What does the comparison of E and F support? The tip may produce a growth-promoting effect that can be transferred.

2. What does the comparison of F, G and H support? The position of treated agar（瓊脂） affects the direction of coleoptile（胚芽鞘） bending.

3. When treated agar（瓊脂） is placed on the **left** side of a coleoptile（胚芽鞘） with its tip removed, compared with the right side, the left side receives more substance X, causing cells on the left to **elongate（延長）** more and the coleoptile（胚芽鞘） to bend to the right side.

B5: independent application

A–D: whether the tip is present. E–F: whether agar（瓊脂） previously touched the tip, with both in the centre. F–G–H: treated agar（瓊脂） position, with the same agar（瓊脂） type. The main comparison changes shading treatment.

Window-side context: initially straighter, then bent towards the light on the right.

The light is on the right, so the left is the shaded side. The tip senses light from one side; auxin（生長素） is more abundant on the shaded left and its growth-promoting effect passes down. Cells on the left elongate（延長） more than those on the right, making the coleoptile（胚芽鞘） bend towards the right-side light.

Cutting creates a wound and removes tissue, so it cannot establish a unique mechanism alone. Dark agar（瓊脂） results support a transferable growth-promoting effect and unequal elongation（延長）. Bending is not caused by light during incubation; the results neither measure auxin（生長素） concentration directly nor identify the substance.

Check and correct

Identify one answer you corrected and the image or comparison supporting the correction. If correct initially, give one supporting piece of evidence.

In one sentence, explain how auxin（生長素）, differential elongation（延長） and bending are related.